

# PROGRAMMABLE INDUSTRIAL ON-OFF TIMER WITH RF REMOTE



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In most of the modern manufacturing and processing industries, there is complete industrial automation through sophisticated hardware and software like programmable logic controller (PLC), distributed control system (DCS), and supervisory control and data acquisition (SCADA). Microcontroller-based embedded systems play major role in industrial automation. One such widely used system is the programmable timer. The major applications of programmable timer are as follows:

1. Initiating the process after desired time
2. Switching on/off the process after predetermined time
3. Providing delay in between processes
4. Applying input to on/off type open-loop control system

Depending upon the requirement of process, one can set the time of programmable timer. As the time period expires, the timer will either trigger or shut the process. Earlier there were mechanical timers that used gear assembly (same as wall clock) and mechanical contacts. But the problem with these was that due to mechanical parts and movements, they were not durable. Electronic timers have become very popular as these have more functionalities and long operating life.

A simple electronic timer can be made using a single IC 555 in monostable mode that can switch on/off the process after desired time. Also, in chain process (where the end of

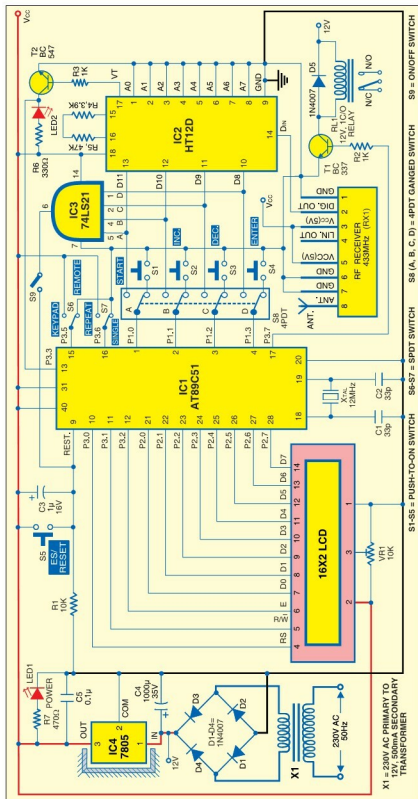


Fig. 1: Circuit diagram of programmable industrial on/off timer